

TECHNICAL PROPOSAL

Annual Maintenance, Technical Support & Enhancement of the NIGST Website

**Client: National Institute of Geo-Informatics Science & Technology (NIGST)
Survey of India, Government of India**

Proposal Type: Technical Proposal / Annual Maintenance Contract (AMC)

Submitted By



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Confidential | Technical Proposal | No Financial Information Included

About the Company

Who we are, what we deliver, and the leadership accountable for this engagement.

We at Source Soft Solutions provide AI, cloud, and data solutions to drive innovation and growth across industries. Our services focus on data modernization, analytics, and cloud infrastructure management, helping businesses unlock new possibilities with AI/ML, data insights, and scalable cloud technologies.

Empowering organizations to thrive in the digital landscape.

Headquartered in Edison, New Jersey, with an office in Dubai and our primary engineering centre in Noida, India, we run long-term maintenance and support contracts on the same platforms we engineer.

What we deliver

Transformative Data Solutions	Robust Cloud Services
Advanced AI Capabilities	Insightful Analytics
Seamless Integration	Custom Software Development
Data Security & Compliance	Training & Support

Leadership team



Alok Dharayan · President

25+ years delivering engineering solutions to complex global problems. Masters, New Jersey Institute of Technology. Sets delivery governance and client accountability across Source Soft Solutions' enterprise and government engagements.



Vijay Konar · Chief Technology Officer

Leads all product and platform engineering at Source Soft Solutions. Deep expertise in full-stack development, AWS cloud architecture and data engineering; owns technical standards, release controls and architecture reviews.



Vikram Sharma · Solutions Lead

Architects client engagements end-to-end – from requirement study and solution design to transition-in and steady-state operations. Background in enterprise consulting and large-scale application maintenance programmes.

00 Document Control

Proposal governance, versioning and submission information.

Document	Technical Proposal — Annual Maintenance, Technical Support & Enhancement of NIGST Website
Client	National Institute of Geo-Informatics Science & Technology (NIGST), Survey of India
Prepared By	Source Soft Solutions Private Limited
Purpose	Technical response describing understanding, methodology, operating model, controls, deliverables and handover approach for the NIGST website AMC.
Source Basis	NIGST Scope of Work and Annexure-1 (Web Application Architecture Overview) supplied with the bid material.
Version	Technical Proposal — AMC (v1.0)
Confidentiality	Confidential — prepared for NIGST technical evaluation and bid submission.

Bid Response Principle. This proposal translates every SOW requirement into an operational workstream, control, process and measurable evidence. The solution is designed to **maintain the existing application architecture** — React SPAs, shared REST API and AWS/Kubernetes infrastructure — rather than introduce an unnecessary re-platforming exercise.

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A structured navigation map for the technical response.

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01 Executive Summary

Maintain the platform. Protect the service. Improve it without destabilising the architecture.



The NIGST Scope of Work requires comprehensive technical maintenance, support and enhancement of the existing website and associated components. The selected firm is expected to keep all website modules stable, proactively monitor uptime and performance, manage incidents and bugs, maintain application and infrastructure health, support databases and backups, maintain SSL/TLS, apply security patches, support **GIGW 3.0** compliance, coordinate security audits, perform in-scope enhancements, operate SMTP, manage content and configuration, maintain SOPs, provide monthly reporting and preserve all digital assets for handover.

Our methodology uses a **closed operational loop**: observe → detect → diagnose → resolve → validate → deploy → document → improve. This keeps routine AMC activity evidence-driven and prevents repeated symptom-level fixes. Because our team is already fluent in the exact stack described in Annexure-1 (React 18.2, Axios, PostgreSQL on Amazon RDS, EKS/Kubernetes), mobilization is fast and low-risk.

Why Source Soft Solutions. Architecture-aware maintenance, a disciplined RCA-first incident model, full SSL/TLS and SMTP lifecycle ownership within the quoted price, GIGW 3.0 alignment, and a complete asset-preservation and handover commitment — delivered by a lean, multidisciplinary team.

02 Understanding of Requirements

The SOW is an operations-and-continuous-improvement engagement, not a greenfield rebuild.

SOW Workstream	Our Understanding	Evidence / Output
Website Operations & Monitoring	Continuous stable operation; proactive monitoring of performance, uptime and availability; preventive action to reduce downtime.	Health checks · monitoring records · preventive actions
Incident Management & Troubleshooting	Incident logging, RCA for recurring issues, fixes within existing architecture and timely resolution.	Incident register · RCA · closure evidence
Bug Fixing & Issue Resolution	Diagnose user-reported and system-detected defects and restore service continuity.	Bug log · fix record · validation
Routine Maintenance & System Health	Periodic application/infrastructure health checks, service restarts, scheduled-job monitoring and diagnostics.	Maintenance checklist · job status · logs
Database Administration & Support	DB health/performance, scripts, connectivity troubleshooting, backup/restoration and integrity.	DB health record · backup/restore evidence
Security, Compliance & Stability	SSL/TLS lifecycle, security patches, GIGW 3.0, audit coordination, vulnerability remediation and closure.	Certificate record · patch log · audit tracker
Enhancement & Modification	Enhance existing modules, workflows, UI, APIs, reports, dashboards, configuration and business logic within existing architecture.	Change request · test evidence · deployment record
Content & Configuration / SMTP	SMTP service, content updates, environment/configuration changes and minor layout/navigation changes.	Config record · content log · SMTP evidence
Runbook & SOP Execution	Runbooks/SOPs maintained and refined based on operational experience.	SOP pack
Reporting & Documentation	Monthly reporting covering performance, maintenance and incidents; audit-ready documentation.	Monthly AMC report
Digital Asset Protection & Handover	Preserve source, DB, APIs, designs, docs, configurations, credentials, licenses; complete handover at closure.	Versioned assets · backup · handover pack

Out-of-scope boundary. New feature development, major system changes, third-party modifications, infrastructure upgrades and licensing-driven changes are treated as separate change requests subject to mutual agreement, effort estimation and approval.

03 Existing Technical Context & Architecture

Architecture-aware maintenance based on the supplied Annexure-1 overview.

Annexure-1 describes NIGST as a multi-application, full-stack system with two independently deployable React Single Page Applications communicating with a shared centralized REST API backend. Communication is secured over HTTPS/TLS on port 443. The documented applications are the public website (**nigstFront**) and the admin dashboard (**nigstdashboard**), deployed on AWS using a containerized Kubernetes (EKS) architecture with Amazon RDS (PostgreSQL).



NIGST WEB APPLICATION — LAYERED ARCHITECTURE (PER ANNEXURE-1)

The maintenance model maps directly to these layers: frontend stability and UI changes; security-control validation; API/service diagnostics; and backend/data operational support. **No unnecessary architecture replacement is proposed.**

04 Proposed AMC Methodology

A practical service lifecycle with clear entry, control and exit conditions.

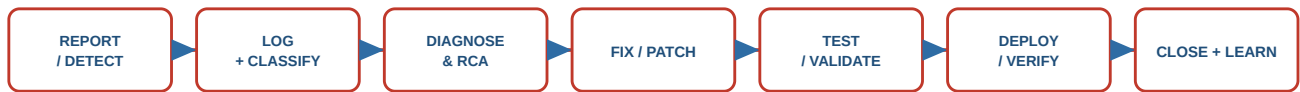


AMC SERVICE LIFECYCLE

Control Area	Method	Primary Deliverable
Baseline & Discovery	Review application, access, repositories, integrations, database, configuration and operational dependencies at mobilization.	Technical baseline + risk/backlog
Monitoring & Prevention	Observe application/service health, logs, scheduled jobs, availability and recurring failure patterns.	Monitoring / maintenance records
Incident Response	Classify, diagnose, perform RCA where required, implement fix and verify restoration.	Incident + RCA + closure evidence
Change Management	Assess impact, develop/configure within scope, test, approve, deploy and document.	Change record + release evidence
Security & Compliance	Maintain certificates, patches, vulnerability remediation, audit coordination and GIGW 3.0 alignment.	Security tracker + remediation evidence
Reporting & Governance	Maintain incident/change/maintenance records and issue periodic reports for review.	Monthly AMC report + action log

05 Incident, Bug & Problem Management

A single controlled workflow for reported issues, detected anomalies and recurring problems.



INCIDENT LIFECYCLE

Incidents are uniquely identified and recorded with source, affected component, impact and status. Diagnosis uses available application/database/infrastructure logs and technical checks. Recurring or high-impact issues are taken through **root cause analysis** and preventive action rather than repeatedly applying short-term fixes.

Stage	Control	Evidence
Intake	Capture issue, source, time, component and symptoms.	Incident ID / ticket
Triage	Assess impact, urgency, dependency and likely scope.	Priority / classification
Diagnosis	Review logs, service health, application/API and database behaviour.	Diagnostic notes
Resolution	Apply approved code/config/data fix within AMC scope.	Fix / change record
Validation	Run functional and regression checks appropriate to risk.	Test evidence
Closure	Verify production recovery and update records/reporting.	Closure note / report

06 Security, Compliance & Stability

Treat certificates, patches, audits and vulnerabilities as a continuous lifecycle.



SECURITY & COMPLIANCE LIFECYCLE

Security Area	Proposed Technical Approach
SSL/TLS	Provision/procure where required, install, configure, renew, validate and maintain website SSL/TLS throughout the AMC period; all related costs are included in the quoted price.
Security Patching	Assess updates, dependencies and patches; test where appropriate; deploy through controlled change and verify service health.
GIGW 3.0	Maintain alignment with the security, accessibility and performance expectations stated in the SOW and track identified gaps and remediation actions.
Security Audits	Coordinate with designated audit teams during scheduled audits; log findings and drive remediation to closure within compliance timelines.
Application Stability	Preserve existing controls and validate authentication, route protection, secure cookies/storage, input sanitisation (DOMPurify) and error handling during every change.

Cost inclusion. As required by the SOW, all costs for procurement, renewal, licensing, installation, configuration, validation and maintenance of the SSL/TLS certificate(s) are included in the quoted AMC price. No additional payment is sought for these services during the contract period.

07 Database, Backup & Recovery

Protect data integrity, performance and recoverability as part of routine operations.



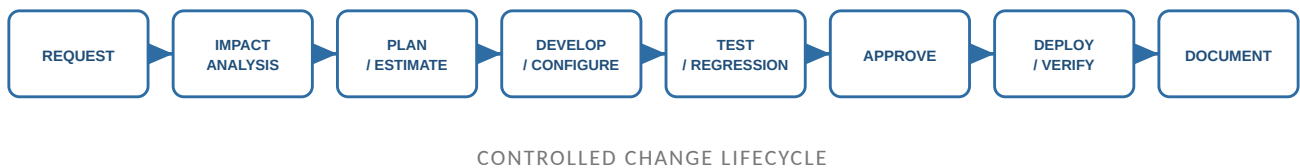
DATABASE OPERATIONS & RECOVERY FLOW

Database support covers routine health and performance monitoring, execution of predefined scripts and maintenance procedures, connectivity troubleshooting, backup and restoration support, and preservation of data integrity, security and optimised performance. The backend runs on Amazon RDS (PostgreSQL), so automated backups and managed patching are leveraged alongside our operational monitoring.

Activity	Operational Control	Evidence
Health Checks	Routine review of database health, connectivity and service behaviour.	DB health record
Performance	Identify abnormal response/connection patterns and investigate bottlenecks within scope.	Performance observations
Backup	Execute/support predefined backup processes and validate completion.	Backup log
Restoration	Support restoration activities when required and verify application/database consistency after recovery.	Restore evidence
Security & Integrity	Protect access, preserve data integrity and document corrective actions.	Access/change record

08 Enhancement & Change Management

Enhance the evolving application without converting AMC into uncontrolled new development.



The SOW explicitly permits enhancement, modification, optimisation and refinement of existing modules, features, workflows, UIs, APIs, reports, dashboards, configurations and business logic **within the existing architecture**. Minor additions are allowed when they extend existing functionality without becoming an entirely new software development project.

In-Scope AMC Enhancement	Separate Change Request
Existing module improvements; UI/navigation refinement; API/report/dashboard updates; workflow or configuration improvements; minor functional additions that extend the existing architecture.	New feature development; major system changes; third-party modifications; infrastructure upgrades; licensing-driven changes or work that materially changes the architecture.

Scope discipline. Every enhancement passes impact analysis and NIGST approval before development, protecting both stability and the boundary between in-scope work and separately-estimated change requests.

09 Content, SMTP & Configuration Management

Support the operational communication and content layer with controlled configuration.



CONTENT & CONFIGURATION FLOW

The SOW requires the selected firm to provision/arrange, configure, operate and maintain SMTP service for application-generated emails including authentication, password reset, notifications and other system-generated communications. Procurement/subscription, renewal, configuration, operation, maintenance and technical support costs are included in the quoted AMC price.

Area	Support Model
SMTP	Configure, operate, maintain, troubleshoot and renew the application email service (authentication, password reset, notifications and other system-generated emails) as required by the SOW.
Content	Support authorised uploads, updates and management of website content.
Configuration	Apply controlled changes to environment variables, configuration files and system settings.
UI / Layout	Minor layout/navigation changes within the existing design framework.

10 SOP, Reporting & Governance

Operational knowledge is converted into repeatable procedures and auditable evidence.



GOVERNANCE & REPORTING CYCLE

Reporting Layer	Indicative Contents
System Performance	Availability/health observations, major alerts, performance trends and actions.
Maintenance Activities	Application, infrastructure, database, certificate, configuration and preventive maintenance activities.
Incident Logs & Resolutions	Incident register, RCA, resolution status, recurring issue actions and closures.
Changes & Deployments	Bug fixes, minor enhancements, deployments, approvals and verification records.
Security	Patch activity, audit findings, vulnerability remediation and closure status.
SOP / Runbook	Operational procedures updated based on experience, recurring issues and approved system changes.

Monthly reporting. As required by the SOW, monthly reports cover system performance, maintenance activities, and incident logs and resolutions, together with proper documentation for audit and reference.

11 Asset Protection, Version Control & Handover

Preserve the complete technical baseline and make exit/transition predictable.

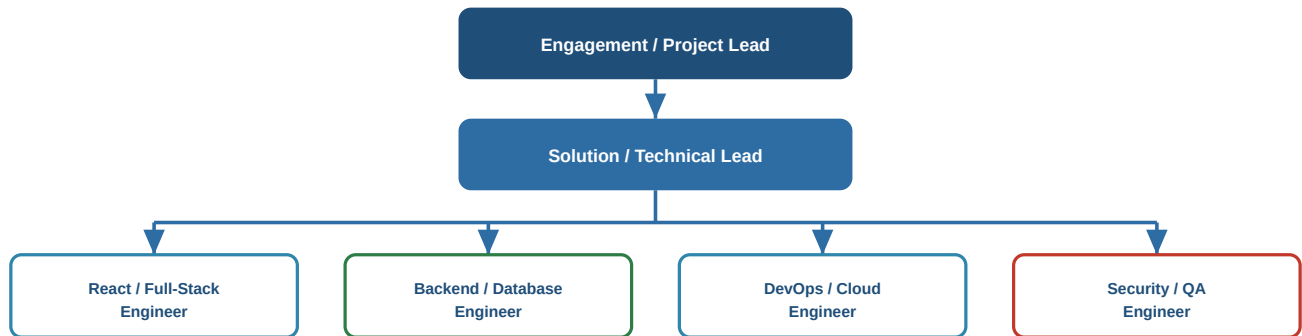


ASSET PRESERVATION & HANDOVER CHAIN

Asset	Protection / Handover Control
Source Code	Maintain current source and version-control history; preserve a usable build/deployment baseline.
Databases	Maintain backups and associated recovery information; preserve data integrity.
APIs & Integrations	Maintain interface/configuration documentation and relevant operational dependencies.
Design & Documentation	Preserve design elements, architecture notes, SOPs, runbooks and technical documentation.
Credentials & Configuration	Protect access information and configuration records through approved access-control processes.
Licenses & Digital Resources	Preserve applicable license records and associated project resources.
Closure / Handover	Hand over IP rights, source code, documentation, credentials, databases and associated project assets in a complete and usable form without restriction or withholding.

12 Delivery Team & Responsibilities

A lean multidisciplinary team with clear accountability across application, database, cloud, security and governance.



AMC DELIVERY TEAM STRUCTURE

Role	Primary Responsibility	Engagement
Engagement / Project Lead	NIGST coordination, governance, reporting, risk/escalation and approvals.	Governance + reviews
Solution / Technical Lead	Architecture stewardship, impact analysis and complex troubleshooting.	On-demand + changes
React / Full-Stack Engineer	Public/admin UI, APIs, workflows, defects and in-scope enhancements.	Core support
Backend / Database Engineer	API services, DB health, scripts, backup/restore and performance support.	Core + escalation
DevOps / Cloud Engineer	Deployment, logs, monitoring, certificates and operational controls.	Core + escalation
Security / QA Engineer	Security remediation, regression, release validation and audit support.	Planned + release/audit

Named personnel and CVs to be inserted from the final bidder resource pool and documentary evidence. This proposal intentionally avoids inventing personnel credentials.

13 Mobilization / Transition-In Plan

Establish the operational baseline before steady-state AMC delivery.



TRANSITION-IN SEQUENCE

Phase	Key Activities	Output
1. Kick-off	Confirm SPOC, escalation path, access, communication channels and operating cadence.	Mobilization checklist
2. Discover	Review architecture, source repositories, application components, DB, APIs, configuration and dependencies.	Technical baseline
3. Baseline	Run initial health/security/operational checks and identify immediate risks.	Baseline report
4. Operationalize	Configure/validate monitoring, incident process, backup routines, SOPs and reporting.	AMC operating baseline
5. Steady State	Run preventive maintenance, incident resolution, security remediation, enhancements and monthly reviews.	Continuous service delivery

14 Quality & Release Controls

Every production change passes a controlled validation path before closure.



CONTROLLED RELEASE PATH

Quality Gate	Control
Technical Review	Check compatibility with existing architecture, configuration and security practices.
Functional Testing	Validate the affected module/function against expected behaviour.
Regression Testing	Check relevant adjacent functionality before release.
Approval	Obtain required NIGST approval/sign-off for deployment where applicable.
Deployment	Use controlled, version-identified release procedures and preserve rollback readiness.
Post-Release Verification	Run smoke checks and monitor for abnormal behaviour after release.
Evidence	Update change/deployment record and associated documentation.

15 Service Levels & Feasibility Assessment

Indicative response priorities and an honest read on delivery risk.

Indicative Priority & Response Model

Severity	Response Approach	Priority
Critical (site down / login broken)	Immediate acknowledgement; continuous work until service restored.	P1
High (module broken, no workaround)	Same business day acknowledgement; prioritised resolution.	P2
Medium (partial issue, workaround exists)	Next business day; scheduled into release cycle.	P3
Low (cosmetic, minor request)	Planned; bundled into routine maintenance/enhancement window.	P4

Final SLA targets (response/resolution times, uptime %) to be confirmed with NIGST and captured in the contract.

Feasibility & Risk Read

Area	Risk	Note
Monitoring, health, routine maintenance	Low	Standard AWS/EKS tooling; fully feasible.
Database support (RDS)	Low	Managed backups/patching reduce effort.
SSL/TLS & SMTP lifecycle	Low	AWS ACM + SES/managed SMTP; cost included in quote.
Bug fixing / enhancements	Medium	Feasible with source + backend access from NIGST.
GIGW 3.0 & audit remediation	Medium	Feasible; remediation effort tracked as it is identified.

Dependencies from NIGST. Timely access to applications, servers, databases, APIs and source repositories; valid licenses for third-party tools; a nominated SPOC; and timely approvals/sign-offs for fixes, deployments and minor enhancements.

16 Conclusion & Value Proposition

A stable, secure and continuously improving NIGST platform with complete operational visibility.

Source Soft Solutions proposes a disciplined AMC model aligned to the supplied NIGST SOW and Annexure-1 architecture. The approach is deliberately practical: maintain the existing platform, detect issues early, resolve them through repeatable RCA and testing, support database and infrastructure health, maintain SSL/TLS and security controls, coordinate audit remediation, provide controlled enhancements, operate SMTP and content/configuration support, and preserve the full technical asset baseline for future continuity.

NIGST Need	Proposed Response
Stable website	Proactive monitoring, health checks, incident response and preventive maintenance.
Responsive support	Structured intake, prioritisation, RCA, tested fixes and closure evidence.
Security & compliance	SSL/TLS lifecycle, security patches, GIGW 3.0 alignment and audit remediation.
Controlled evolution	Enhancements inside the existing architecture with change and release controls.
Operational transparency	Incident/change records, SOPs and monthly performance/maintenance reporting.
Continuity & ownership	Version control, backup, asset preservation and complete handover.

Proposal design note. This technical response uses process flows and architecture visuals rather than decorative application mock-screens because the supplied SOW is an AMC/operations engagement. Visuals are used where they improve technical comprehension: architecture, incident lifecycle, security lifecycle, change lifecycle, database recovery and handover.

A Annexure A — Requirement Traceability Matrix

Direct mapping of SOW requirements to proposal sections.

#	SOW Requirement	Proposal Section
1	Website Operations & Monitoring	02, 04
2	Incident Management & Troubleshooting	02, 05
3	Bug Fixing & Issue Resolution	02, 05
4	Routine Maintenance & System Health	02, 04
5	Database Administration & Support	02, 07
6	Security, Compliance & Stability	02, 06
7	Enhancement & Modification	02, 08
8	Content & Configuration / SMTP	02, 09
9	Runbook & SOP Execution	02, 10
10	Reporting & Documentation	02, 10
11	Digital Asset Protection & Handover	02, 11
—	Out-of-Scope: separate change request for activities beyond defined AMC scope	02, 08
—	Annexure-1 Web Application Architecture Overview	03

Coverage statement. Every requirement in the NIGST Scope of Work, including the out-of-scope boundary and Annexure-1 architecture, is addressed within this proposal.

B Annexure B — Company & Contact Details

Source Soft Solutions — contact and office information.

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End of Technical Proposal

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